

The Expanding Universe Model Does Not Hold

Evidence from Observed Geometry,
Visibility Thresholds, and Frequency Physics

Annex to the Harmonic Structures in Natural Systems Series

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Abstract

The standard cosmological model holds that the universe is physically expanding: space itself stretches, carrying galaxies apart, and observed redshift in distant light is the velocity signature of that recession. This annex argues that the model directly contradicts observable, precisely measured physical reality on multiple independent grounds.

Four converging lines of evidence are presented. First, the geometric record: constellations and deep-field celestial structures have been mapped with sub-arcsecond precision across millennia and show no angular drift consistent with expansion at any scale, including regions the standard model designates as expansion-dominated. Second, Hubble's original interpretive step: the linear redshift-distance relationship he identified in 1929 was real, but the velocity interpretation applied to it was the best available inference at the time, not a validated physical derivation, and the data accumulated since does not confirm it. Third, the redshift reinterpretation: under the curvature-frequency identity $f = c/r$, cosmological redshift is a natural perimeter attenuation effect at the edge of any detector's visibility window, not a recession velocity signal. Fourth, the conflation of separately measured parameters: the cosmological energy budget percentages attributed to dark matter and dark energy are model-fitted quantities, not independent measurements, and they are being reported with the same empirical status as directly observed local density ratios.

The conclusion is direct: the universe is not expanding. Expansion was a geometric misreading of a visibility threshold effect, compounded by an interpretive step that was never independently validated and a parameter conflation that has gone largely unnamed in the literature.

This annex builds on three prior publications in the Harmonic Structures in Natural Systems series and should be read in conjunction with them for full derivations.

1. The Internal Contradiction of the Expansion Model

The expanding universe model rests on a specific observational claim: that distant galaxies are receding from us in all directions, and that this recession scales linearly with distance. To explain this without placing our location at a physically privileged center, the model requires that space itself is stretching, not that objects are moving through static space.

This generates an immediate and unresolved internal contradiction. If space itself is stretching uniformly, the geometric relationships between all objects embedded in that space should drift over time. The stars we observe tonight should not occupy the same angular positions relative to one another as the stars Hipparchus catalogued in 129 BCE, or as the stars encoded on the ceiling of Senenmut's tomb circa 1470 BCE. They do not.

The standard model's response is a distinction between bound systems, where local gravity overcomes expansion, and unbound space, where expansion proceeds freely. This distinction is not derived from any first principle of the model itself. It is applied after the fact to reconcile the theory with observation. Furthermore, as Section 3 demonstrates in detail, the partition fails on its own terms: even in regions the model classifies as unbound, far beyond our galactic neighborhood, deep-field surveys reveal coherent, stable angular geometry between objects. The grid is not stretching anywhere we can actually measure it.

2. Hubble's Observation and the Interpretive Step That Was Never Validated

Edwin Hubble's 1929 observation was precise and it stands: redshift in the light from distant galaxies scales linearly with distance. That is a real measurement. It has been confirmed and refined repeatedly in the nearly one hundred years since.

The interpretation applied to that observation, that the redshift encodes the recession velocity of the source, was the most reasonable inference available given the conceptual tools and the data that existed in 1929. It was not an error given what Hubble had. The Friedmann equations from general relativity permitted an expanding metric solution, and the redshift data appeared to fit. The model was adopted because it worked, not because competing explanations were formally ruled out.

What has not been adequately examined since is whether that interpretive step, redshift as recession velocity, was ever independently validated as the unique physical explanation for the observed signal. It was not. The data Hubble did not have in 1929 is now in hand. It does not confirm the velocity interpretation.

2.1 The Doppler Analogy and Its Limits

The velocity interpretation of redshift borrows from the Doppler effect for sound, where a moving source compresses or stretches waves relative to a stationary observer, producing a frequency shift that

encodes source velocity. It is worth noting at the outset that an alternative mechanism exists that produces the identical linear redshift-distance relationship without any recession velocity. That mechanism is introduced in Section 2.3 and developed fully in Section 4.

Doppler for sound is fully validated because every variable in the transfer function is independently measurable: the emission frequency of the source at rest, the propagation speed of the medium, and the source velocity confirmed by separate instruments. Nothing is assumed. Nothing is fitted. The relationship holds at every scale it has been tested precisely because all variables are independently traceable.

Cosmological redshift has none of those conditions. The emission frequency of a distant galaxy assumes identical atomic physics across billions of light years. The propagation properties of the intergalactic medium at those distances are not independently characterized. The distance to the source is itself partially derived from the redshift, introducing circularity. And the source velocity has no independent measurement whatsoever. The redshift is the only observable. Everything else is derived from it under the expansion assumption.

In any engineering context, a calculation run with model-fitted rather than independently measured variables would not be accepted as a valid measurement. The output carries no traceable uncertainty bound. That standard has not been applied consistently to cosmological redshift.

2.2 The Two Variables Being Conflated

The core conflation in the redshift interpretation is between two physically distinct quantities being treated as if one directly encodes the other. Variable 1 is the frequency response of the detector or observer: its window, its threshold, its characteristic propagation speed for the stratum it operates in. Variable 2 is the velocity of the emitting object, a kinematic property of the source entirely independent of the detector.

Claiming that Variable 1 gives you Variable 2 requires a validated transfer function between them. For sound Doppler that transfer function exists and has been built from independently measured components at multiple scales. For cosmological redshift it has not. The transfer function was borrowed from the sound case and assumed to apply at cosmological scales where none of the intermediate validation steps were performed.

The application of Doppler to light follows a chain of decreasing variable control: radar and satellites where all variables are independently measured and the physics is confirmed; local stars where assumptions are small and results are reasonably reliable; binary stars where radial velocity is still partly derived from the shift being interpreted; galactic rotation curves where the medium is uncharacterized and the dark matter inference sits on top of an unverified Doppler application; and finally cosmological redshift where the system is completely underdetermined and all unknowns except the shift itself are model-fitted. Each step in that chain inherited the assumption from the previous step without re-validating it at the new scale.

2.3 What the Same Linear Relationship Actually Requires

The curvature-frequency identity $f = c/r$, derived in the companion publications of this series, produces the same linear redshift-distance relationship Hubble measured from a completely different mechanism. As propagation distance r increases, detected frequency decreases proportionally, placing distant signals progressively toward the red boundary of any detector's visibility window. No recession velocity required. No expanding metric required.

Hubble's plot does not distinguish between these two mechanisms. A linear redshift-distance relationship is predicted by both. The data that was missing in 1929 to tell them apart is now in hand: three thousand years of stable geometric record, microarcsecond astrometry across one billion stars, deep-field imaging to redshifts above $z = 10$, and gravitational wave detection across multiple frequency bands. None of it produces the angular drift signature that a genuinely stretching coordinate grid would leave in the observational record.

3. The Constellation Argument: A Fixed Geometric Grid

3.1 The Historical Record

The oldest surviving star catalogue is attributed to Hipparchus of Nicaea circa 129 BCE, who recorded the positions of approximately 850 stars with arc-minute precision using only naked-eye instruments. His catalogue was later incorporated into Ptolemy's *Almagest* circa 150 CE, which remained the precision reference for celestial positions for over a millennium.

Before Hipparchus, the Babylonian MUL.APIN tablets circa 1200 BCE documented the positions and seasonal risings of stars and constellations across the full ecliptic. Egyptian astronomical ceilings, including the Senenmut tomb ceiling circa 1470 BCE and the Ramesseum astronomical ceiling, encode star positions consistent with the same geometric relationships we map today.

Across more than three thousand years of independent observation, using instruments ranging from naked-eye sighting tubes to modern charge-coupled device arrays, the angular separations between stars within constellations have remained constant to the precision of measurement. Orion's belt stars, Alnitak, Alnilam, and Mintaka, subtend the same arc they did when Egyptian astronomers first encoded them. The Big Dipper's characteristic geometry, used to locate Polaris, has not drifted. Modern astrometry from the Hipparcos satellite and the Gaia mission measures stellar positions to microarcsecond accuracy. No systematic angular expansion is detected in the local stellar field.

3.2 Proper Motion Is Not Expansion

The standard model acknowledges stellar proper motion: individual stars do have small transverse velocities relative to the Sun, producing measurable angular drift over centuries. Barnard's Star, the fastest-moving star in the sky at 10.3 arcseconds per year, is the canonical example. The Gaia mission has produced proper motion measurements for over 1.4 billion stars.

Proper motion is not expansion. It is individual stellar kinematics within a fixed geometric framework: stars moving through space, not space stretching beneath them. If the underlying coordinate grid were expanding, the signature would be a uniform, direction-independent increase in angular separation between all objects at a given distance scale, systematically scaling with distance. No such signature is observed. What we observe is the opposite: a stable grid with individual objects executing bounded, non-systematic motions within it.

3.3 The Deep Field: Expansion-Designated Zones Show Fixed Geometry

The standard model predicts that beyond the scale of gravitationally bound structures, in the voids between galaxy clusters, the expansion signal should dominate and geometric drift should accumulate over time. The evidence from deep-field surveys contradicts this.

The Coma Cluster, located approximately 321 million light-years from Earth, sits well beyond any definition of locally bound space. It contains over 1,000 identified galaxies. Repeated imaging over decades shows no measurable angular drift between member galaxies inconsistent with their known internal velocity dispersion. The geometric relationships between Coma Cluster members are stable.

The Virgo Supercluster, spanning approximately 110 million light-years, has been mapped extensively through the 2dF Galaxy Redshift Survey and the Sloan Digital Sky Survey. Its internal geometry shows coherent filamentary and void structures that have not blurred or stretched on observable timescales. The Virgo Cluster core itself, at approximately 54 million light-years, shows no angular expansion signature.

The Bullet Cluster, at approximately 3.8 billion light-years, is one of the most studied objects in deep-field cosmology. Its geometry, including the separation between the gas cloud and the inferred gravitational center, has been imaged repeatedly across independent observatories. The angular relationships are stable and reproducible.

At the largest scales, the Hubble Ultra Deep Field and the James Webb Space Telescope deep fields show galaxies at redshifts above $z = 10$, corresponding to lookback times of over 13 billion years. These objects are resolved as coherent, geometrically distinct structures, not smeared or angularly drifting signals. Their internal geometry is measurable.

If the expansion prediction were correct at the rate required by Hubble's Law (H_0 approximately 70 km/s/Mpc), the angular separations between objects in these fields should show systematic, distance-proportional drift on observational timescales accessible to precision instruments. They do not. The grid is fixed.

3.4 What the Geometric Record Requires

The geometric record, spanning three millennia of local observation and decades of deep-field survey imaging, is consistent with one physical model: the underlying coordinate grid of the universe is not expanding. Objects in it have proper motions, and distant objects exhibit redshift. Neither observation requires or implies a stretching coordinate system.

For the expansion model to remain viable in the face of this record, it must identify a mechanism that suppresses every observable expansion signature across every scale we can measure, from local stellar fields to clusters 3.8 billion light-years away, while preserving redshift as a recession signal. That burden has not been met. Invoking gravitational binding does not explain the stability of unbound deep-field structures. It shifts the question without answering it.

The constellations, the deep-field clusters, and the JWST ultra-deep fields are not anomalies to be explained away. They are the primary data. Any model of the universe must account for them as its baseline, not its exception.

4. Cosmological Redshift as Perimeter Attenuation

The full derivation of the curvature-frequency identity $f = c/r$ is presented in the companion publications *Light as a Visibility Threshold* (Johnson, 2025) and *The c-Constant Framework, Annex 1.2* (Johnson, 2025). Readers seeking the complete treatment should consult those publications directly.

4.1 The Visibility Boundary

For any detector with a minimum detectable frequency $f_{\text{threshold}}$, there exists a maximum observable radius $r_{\text{visible}} = c / f_{\text{threshold}}$. As an oscillating signal propagates outward and r increases, its effective detected frequency decreases. When r approaches r_{visible} , the signal frequency approaches the red end of the detection window. Beyond r_{visible} the signal exits the detection band entirely. It has not stopped. It has crossed the visibility boundary.

$$f = c / r \quad r_{\text{visible}} = c / f_{\text{threshold}}$$

4.2 Why the Farthest Objects Look Red

Cosmological redshift, under this reading, is not a velocity signature. It is a perimeter effect: the farthest observable objects are those whose signal frequency has been most attenuated by propagation distance, placing them at the long-wavelength edge of the detection window. Red is simply the last frequency band before a signal drops below threshold.

This requires no expanding space, no recession velocity, and no dark energy to drive acceleration. It requires only that detection is threshold-dependent, which is confirmed directly by every field of measurement science that works with wavelength-sensitive instruments.

4.3 The Biological and Engineering Confirmation

The threshold-dependence of detection is not a theoretical claim. It is the working basis of multiple applied fields that deal with it daily.

Ophthalmologists and optometrists map the human visibility window with precision: the human eye detects wavelengths from approximately 380 nm to 700 nm. Signals outside this band are not seen. They are not absent.

Infrared cameras detect thermal radiation invisible to the human eye. The same room that appears uniformly dark at night is a rich field of radiating objects to an infrared sensor. The photons do not change. The detector window changes.

Laser physicists work daily with coherence requirements for visibility. A laser beam remains detectable at great distances precisely because its coherence maintains the oscillation within the detector window far longer than an incoherent source of the same power. Radio astronomers, X-ray observatories, and gravitational wave detectors each operate in distinct frequency strata. All Λ^2 electromagnetic modes propagate at $c = 3 \times 10^8$ m/s. This consistency is a property of the stratum, not a universal constant independent of detection.

The universe does not change as we change instruments. Our visibility window does. Cosmological redshift is the same effect operating at astronomical distances.

5. Dark Matter, Dark Energy, and the Parameter Conflation

This section draws on *The Invisible Spectrum: Dark Matter, Dark Energy, and Black Holes* (Johnson, 2025). Full derivations and testable predictions are contained in that publication.

5.1 Two Parameters Being Reported as One

The standard model reports the universe's composition as approximately 5 percent visible matter, 27 percent dark matter, and 68 percent dark energy. These percentages are presented as measurements. They are not. Two entirely separate parameters with different epistemic status are being conflated.

Parameter 1 is the local density ratio: how much non-luminous mass exists relative to visible mass in observed regions. This comes from galactic rotation curves, gravitational lensing, and cluster dynamics. It is a real measurement of mass distribution within bounded systems. It does not require and does not imply an expanding universe.

Parameter 2 is the cosmological energy budget: the global percentage breakdown derived by fitting the Lambda-CDM model to the CMB power spectrum and supernova redshift data under the expansion assumption. These percentages are what the model requires to produce the observed CMB acoustic peaks given an expanding metric. They are fit parameters, not independent measurements. Change the metric assumption and they change entirely.

Treating Parameter 2 as if it has the same empirical status as Parameter 1 is the conflation. The 27 percent figure is not a measured fact about the universe. It is the value the model requires under the expansion assumption.

5.2 The Rotation Curve Problem and Its Inherited Assumption

The galactic rotation curve argument, which is the empirical basis for dark matter at local scales, sits in the middle of the Doppler validation chain described in Section 2. We measure spectral line shifts from stars at different radii in a galaxy, interpret them as orbital velocities via Doppler, find the velocities do not match the visible mass distribution, and infer dark matter.

But if the frequency shift is not purely encoding velocity, and if propagation path properties or threshold effects are contributing at galactic scales, then the rotation curve velocities are not clean velocity measurements. The dark matter inference sits on top of an application of Doppler that has not been independently validated at galactic scales. The dark matter argument and the expansion argument share the same unverified premise at their base. They are not two independent lines of evidence for the standard model. They are two conclusions drawn from the same unvalidated premise.

5.3 The Dark Energy Percentage Cannot Be Stable

The model requires dark energy's proportional share to increase continuously as the universe expands, maintaining constant energy density against a growing volume. This means the 68 percent figure is not a stable measured quantity. It is a moving target being reported as a fixed fact.

Furthermore, the predicted dilution of dark matter density over cosmic time, which the model also requires, has never been measured. The observational baseline we have is decades. The timescale over which dilution would be detectable is orders of magnitude longer. The change is entirely model-derived, not observed.

Under the alternative reading presented in this annex and in *The Invisible Spectrum*, dark matter and dark energy are frequency bands in the spacetime oscillation spectrum outside the current human and instrumental detection window. They require no new particles, no expanding space, and no unstable percentage bookkeeping.

6. The Local-Cosmic Partition Tracks the Data Rather Than Predicting It

The standard model's response to the constellation stability argument is the local-cosmic partition: space does not expand inside gravitationally bound systems because local binding forces are stronger than the expansion rate.

The Friedmann equations, which are the mathematical basis of the expansion model, describe a uniform expansion of the metric. They do not contain a term that switches expansion off below a certain mass density. The partition is a physical intuition applied separately to reconcile the theory with observation.

The partition also relies on a scale boundary that the model cannot cleanly specify. At what mass density does gravitational binding overcome expansion? The answer shifts depending on which observed stability is being explained. The Milky Way is bound. The Local Group is bound. The Virgo Supercluster turns out also to be bound under the model's accounting. The Laniakea Supercluster, spanning 500 million light-years, is the current frontier. Each time an observed stability is documented, the bound-system boundary is revised outward.

Even in the regions the model definitively classifies as unbound, the angular geometry of observed structures is stable and reproducible on observational timescales. The model predicts drift in these regions. The data shows none. A boundary that moves to accommodate each new observation rather than predicting it in advance is tracking the data, not explaining it.

7. Synthesis: What the Evidence Actually Shows

The converging lines of evidence presented here all point in the same direction.

The geometric record of three thousand years, from Babylonian cuneiform tablets to the Gaia mission's catalogue of over one billion stars, shows a universe whose angular geometry is stable at every scale we can measure. Proper motions exist, but the underlying grid does not drift.

Deep-field observations of the Coma Cluster, the Virgo Supercluster, the Bullet Cluster, and the James Webb Space Telescope ultra-deep fields show coherent, stable, measurable geometry in regions the expansion model designates as expansion-dominated. No angular drift consistent with the predicted Hubble rate is observed.

Hubble's 1929 observation was real and precise. The velocity interpretation applied to it was never independently validated. The data that was missing in 1929 is now in hand. It does not support that interpretation.

Cosmological redshift is fully explained by the visibility threshold effect derived from $f = c/r$ without any recession velocity and without any stretching coordinate system. The detection-threshold nature of visibility is the daily working reality of ophthalmology, laser physics, radio astronomy, infrared imaging, and gravitational wave detection. Every practitioner in those fields works with the principle that absence from the detection window does not imply physical absence.

The cosmological energy budget percentages are fit parameters under the expansion assumption, not independent measurements. They are being reported with the same empirical status as directly observed local density ratios. That conflation needs to be named.

The local-cosmic partition, the model's primary defense against the constellation argument, is a boundary that has been revised outward each time a new stability is observed. It is not a prediction. It is an accommodation.

8. Conclusion

The universe is not expanding.

This conclusion follows directly from the observational record when the recession velocity interpretation of redshift is not assumed at the outset.

The constellations have not drifted. The Coma Cluster has not expanded. The Bullet Cluster geometry is stable. The James Webb deep fields show resolved, coherent structures at lookback times of over 13 billion years. These are not anomalies to be explained away by a local-cosmic partition.

Cosmological redshift is a visibility threshold effect. The curvature-frequency identity $f = c/r$ produces a natural perimeter attenuation that places the farthest detectable signals at the red edge of any detector's window. This is the same physics that makes a candle appear red at distance, that makes a

laser remain visible longer than an incoherent source, and that makes infrared cameras see what human eyes cannot. The physics is known and confirmed across multiple applied fields. Its cosmological application has been available but overlooked.

Hubble made the best call available with what existed in 1929. The interpretive step he took was reasonable under the constraints of his moment. It was not mathematically validated as the unique explanation for the data, and the century of observation that followed has not supplied that validation. What it has supplied is a geometric record detailed enough to read without prior assumptions.

That record points to a universe whose coordinate grid is fixed, whose redshift is a threshold effect, and whose dark constituents are frequency bands outside our current detection window. Not a universe that is expanding, accelerating, and composed mostly of substances we have never directly detected.

"We do not know why we are born into the world, but we can try to find out what sort of a world it is — at least in its physical aspects."

Edwin Hubble, as quoted in Edwin Hubble: Mariner of the Nebulae (Gale E. Christianson, 1996), p. 183.

"With increasing distance, our knowledge fades, and fades rapidly. Eventually, we reach the dim boundary — the utmost limits of our telescopes. There, we measure shadows, and we search among ghostly errors of measurement for landmarks that are scarcely more substantial."

Edwin Hubble, The Realm of the Nebulae (1936).

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